The Spherical Model: Universe Core, Gravitation, and Intrinsic Expansion

Author: Andrew Robus

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Starting Point: The Missing Gravitation in the Milky Way

We start from the observation that stars in the outer regions of the Milky Way move faster than expected based on the gravitational effect calculated from visible matter. The established explanation for this is dark matter. In our model, we investigate instead whether a very massive, distant universe core could generate a portion of this additional gravitational effect. For this, we performed an initial rough estimate calculation. Our assumptions:

- Distance between the universe core and the Milky Way: 46 billion light-years. (The core could also be much further away, which would mean the masses of the core and the universe would also be larger. In my opinion, 46 billion light-years is the minimum distance to the core, which is why I use this number as a basis)
- Visible mass of the Milky Way: approximately 6.24 times 10 to the 10th power solar masses.
- The additionally required gravitational effect was approximated in a heavily simplified calculation by a missing mass of approximately 4.11 times 10 to the 10th power solar masses.
- We investigated what mass a distant central body would need to have in order to generate a corresponding additional acceleration via its spatially variable gravitational effect.

The preliminary result: Core mass approximately equal to 1.05 times 10 to the 29th power solar masses. Compared to the estimated mass of the observable universe of approximately 1.5 times 10 to the 24th power solar masses, this yields: Core mass approximately equal to 70,000 times the mass of the observable universe.

First Rough Calculation

This is explicitly not yet conclusive proof. The calculation uses a simplified Newtonian approximation and does not correctly treat the galactic mass distribution. In particular, the entire visible mass of the Milky Way is not concentrated within the solar orbit. Furthermore, an unresolved shape problem arises: A single distant core generates a specific spatial acceleration distribution via its gravitational or tidal effect. This does not necessarily have to match the distribution actually required within a galaxy. We therefore want to investigate whether two further effects might compensate for or alter this problem:

- Cosmic expansion: The dynamic evolution of spatial distances could counteract the gravitational effect of the central core and thus influence the resulting motion of matter.
- The proper rotation of the galaxy: Rotation and the associated dynamic effects could also influence how the motion of stars is shaped under the influence of additional gravitation.

- We want to examine whether either of these effects alone or both together could generate the required acceleration distribution. It is therefore still open whether this can solve the shape problem and reproduce the observed rotation curves.

Basic Assumption About the Nature of Particles

A fundamental assumption of our hypothesis concerns the nature of elementary particles. We suspect that all particles are fundamentally different states of photons or energy. The various types of particles do not arise from fundamentally different building blocks, but rather from different states and properties of the same energetic foundation. These include in particular:

- Energy,
- Spin,
- Polarization,
- Direction of motion,
- Intrinsic states of motion,
- And other potentially still unknown properties.

The Hypothesis of Intrinsic Motion

According to established physics, a photon moves locally at the speed of light and has no rest mass. Our hypothesis goes further: The energy of a photon is conserved, but can be converted into another state of motion through an extreme physical process. We distinguish between two idealized states:

1. Predominant forward motion: The energy moves directionally through space. The particle appears as a massless photon.
2. Intrinsic motion: Part of the motion structure is converted into an internal or intrinsic motion. The particle therefore no longer moves exclusively as a whole through space, but possesses a different motion structure that appears from the outside as rest mass.

The idea is that the energy is still fully present and moves at the speed of light, but in a different way. As an intuitive image, one can imagine a person who is either running forward or running in place. The energy is used differently, although the total energy is conserved. A possible model would be an extremely curved motion structure in which a particle moves, so to speak, on a closed path and thereby forms an intrinsic form of motion. Such an entity could be figuratively compared to a dog chasing its own tail: the motion still takes place, but no longer leads to a corresponding forward movement of the entire entity. Depending on the configuration of this internal motion structure, different particles could emerge, such as electrons or quarks. Quantum mechanical spin could play an important role here. However, whether it can actually be traced back to such an intrinsic motion structure remains unclear. Important Distinction: Established quantum physics describes spin as intrinsic angular momentum, not as a proven mechanical rotation of a small photon. Furthermore, it is not yet known that electrons or quarks are merely photons in other states of motion. This is a fundamental proprietary hypothesis of our model that would first need to be mathematically and experimentally substantiated.

The Composition and Formation of the Universe Core

Based on the assumed magnitude of the central core, we discussed a possible explanation for its formation. Our approach assumes that the universe was originally a connected, extremely dense, and high-energy sphere. At this point, there was no spatial separation between an inner core and an outer matter region. The history of formation could have proceeded as follows:

1. In a superordinate universe, a star collapses into a black hole and shields itself largely from its universe.
2. Since it can now hardly release energy to the outside, but the total mass of the black hole continues to collapse via gravity and act on its own center (vibrations, for example, and other effects), its energy increases more and more.
3. The early universe is in an extremely high-energy state.
4. Particle-antiparticle pairs are continuously created through pair production.
5. Particles and antiparticles can annihilate, producing photons from which new particle-antiparticle pairs can in turn arise.
6. Under extreme conditions, a hitherto unknown mechanism could lead antiparticles to increasingly organize themselves spatially and concentrate in the center (for example, when the black hole has reached its maximum possible energy density).
7. In particular, positrons, antiprotons, antineutrons, and other antiparticles could aggregate into a common, massive antimatter core.
8. Ordinary matter predominantly remains in the outer region and is spatially separated from the emerging antimatter core.

The crucial thought is that not exclusively positrons gather in the center, but rather the antiparticles of the matter later present in the outer universe.

Electrical Neutrality of the Antimatter Core

This extension is important because a core consisting solely of positrons would possess an enormous positive electrical charge. An antimatter core, on the other hand, can fundamentally be electrically neutral if its positive and negative charge fractions balance each other out. For example:

- Positrons possess a positive electrical charge.
- Antiprotons possess a negative electrical charge.
- Antineutrons are electrically neutral.

An antiproton consists of two anti-up quarks and one anti-down quark. Its total charge is minus 1. A positron possesses a charge of plus 1. An antimatter system with a corresponding particle ratio can therefore possess a total charge of zero. The strong force can also bind antiprotons and antineutrons into anti-nuclei. Positronium or positrons can exist in bound states around such anti-nuclei. This initially eliminates the problem that an exclusively positively charged core would be extremely difficult to keep stable due to its own electrical repulsion. However, electrical neutrality depends on the actual composition and charge distribution. It does not follow automatically from the fact that the core consists of antimatter.

From the Antimatter Core to the Universal Core Quark

Another possibility is that the central antimatter core collapses further under its enormous self-gravity. In doing so, it could reach an extremely concentrated energy state that would be described in classical general relativity as a singularity. In our particle hypothesis, this state could potentially be interpreted as a single, extremely high-energy particle. We provisionally refer to this hypothetical state as: Universal Core Quark. The assumed energy of this state roughly corresponds to the energy of the original antimatter portion, provisionally 70,000 observable universes. The particle could possess an extremely small spatial extent and possibly be structured like a photon or a quark. We must keep two possibilities open:

- The antimatter core actually collapses into an extremely concentrated, point-like energy state.
- The antimatter core remains as a spatially extended system whose actual size and internal structure are hitherto unknown.

In classical general relativity, the singularity is initially a mathematical boundary of description. Whether it corresponds to a physical particle, a point-like energy state, or another structure is unresolved. The Universal Core Quark is therefore a possible interpretation within our own particle hypothesis and not an established description of a black hole singularity. Important: The entire scenario is a speculative assumption. In particular, no physical mechanism is currently known that would reliably produce such a macroscopic separation and concentration of matter and antimatter.

The Mass of the Outer Universe

If we assume that the central core consists of the antiparticles of the matter later present in the outer universe, an interesting consequence arises: The energy of the central antimatter portion should approximately equal the energy of the outer matter portion. We therefore provisionally set: Core mass approximately equal to 70,000 times the mass of the observable universe. Outer mass approximately equal to 70,000 times the mass of the observable universe. This results in for the entire system: Total mass approximately equal to 140,000 times the mass of the observable universe.

This is our provisional mass assumption for the spherical model. The assumed mass equality follows from the hypothesis that originally corresponding amounts of matter and antimatter were formed and subsequently spatially separated. However, it is not yet a proven consequence of particle-antiparticle symmetry. We must later account for how much energy is actually retained through annihilation and other processes and what mass-energy balance the entire system possesses. The energy of the hypothetical Universal Core Quark would already be included in the share of the central antimatter core. It must not be added additionally to the total energy. The previous calculation thus provides a provisional order of magnitude, not a secured total energy balance.

The Universe as a Black Hole

A central assumption of our model is that an ordinary black hole does not exist somewhere within the universe, but rather that the entire universe itself is a black hole. The outer Schwarzschild region or the outer causal boundary thus belongs to the entire universe. The central antimatter core or the possible Universal Core Quark is located within this structure. The outer ordinary matter is also located within the black hole and forms the region that we perceive as our universe. We distinguish

between two levels of observation: The external perspective: The entire system possesses a specific outer boundary. In our hypothesis, its size initially remains unchanged. The internal evolution should not necessarily lead to growth of the outer Schwarzschild radius. The internal perspective: We are located inside the black hole and observe the spatial relationships between the central energy state and the outer matter. The internal spacetime can evolve dynamically in the process, even if the outer boundary of the overall system remains unchanged. The outer Schwarzschild geometry and the internal spacetime evolution must therefore be investigated separately. However, an unchanged outer mass and an unchanged outer Schwarzschild radius are not yet an automatically fulfilled property of the model. They must be compatible with a suitable relativistic solution.

The Intrinsic Expansion of the Inner Universe

This is one of the central assumptions of the spherical model. We assume that the central antimatter core or the Universal Core Quark influences the internal spacetime geometry through its extreme energy concentration. The resulting change in spatial geometry could lead to the spatial distances between the center and the outer matter increasing increasingly. We explicitly do not want to assume an ordinary explosion in which galaxies fly through static space at specific speeds. The image of a layer of matter flying radially away from the core, thereby increasing the radius in which our layer of matter lies, is also only an illustration of the observable effect. The actual hypothesis states: Spacetime itself changes its internal geometry. Matter does not have to move through static space in the same way as would be the case in an ordinary explosion. From our perspective, it looks as though the central core is moving further and further away from us. In the underlying concept, however, an ordinary movement of the core through space is not necessarily responsible for this. Rather, the spatial distance between the central energy state and our matter region changes through the intrinsic evolution of spacetime. As a result, the center could appear further and further away to us, even though the outer size of the entire black hole remains unchanged.

The Role of the Universal Core Quark

In this model, the Universal Core Quark would not simply be an ordinary particle flying through space. Its extreme energy concentration would determine or at least significantly influence the internal spacetime geometry. The central conjecture states: An extremely concentrated energy state in the center could induce a spherically symmetric change in the internal spacetime, causing the spatial distances within the black hole to increase. Whether such an effect can actually follow from the energy-momentum distribution of the central state is still unresolved. In particular, it must be examined whether ordinary gravitation is sufficient for this or whether the hypothesis requires additional physical properties of the central state.

Long-Term Stability

The internal expansion must remain stable over very long periods of time. A later reversal of expansion or a renewed collapse would theoretically not be fundamentally ruled out. However, our model must initially enable a dynamic evolution in which the distances between the core and outer matter increase over cosmological time scales. Matter and antimatter thereby remain spatially separated. A later complete annihilation is not presupposed.

What the Hubble Relationship Looks Like in Our Spherical Model

The observed Hubble relationship is: v = H times r. It describes that the increase in distance between two cosmic reference points is proportional to their distance. For our model, this means: The further away a region is from the center, the faster its physical distance to the center increases. The corresponding geometric description is: R(t) = a(t) times r. Where:

- r: a fixed coordinate within our sphere,
- a(t): a time-dependent expansion factor,
- R(t): the actual physical distance from the center. Mathematically, this follows: Derivative of R with respect to t = (derivative of a with respect to t divided by a) times R. This results in: H(t) = (derivative of a with respect to t) divided by a(t) and thus: Derivative of R with respect to t = H(t) times R.

This equation describes spatial expansion. In our model, this expansion is not meant as ordinary movement of galaxies through static space. It describes the increase in physical distances within dynamic spacetime. The image of a radial distance from the core serves merely for illustration. The mathematical relationship is initially a kinematic description. It does not yet prove that the central energy state can dynamically generate the corresponding expansion.

Our First Hubble Calculation

We subsequently investigated whether the geometry of our spherical model can reproduce the Hubble relationship. For this, we used the assumed distance of the Milky Way from the core of 46 billion light-years. As a comparison value, we used today's Hubble constant of approximately 70 km/s/Mpc. The assumed distance is: R approximately equal to 14,104 Mpc. This results in: 70 times 14,104 = 987,280 km/s. Converted: Derivative of R approximately equal to 3.55 billion km/h.

This number describes the instantaneous increase in physical distance at 46 billion light-years if the expansion rate corresponds to the Hubble constant. It is not a local speed at which the Milky Way flies through space.

What We Have Actually Shown With This

We have shown that a spherically symmetric expansion with spatial extent increasing proportionally to distance can mathematically reproduce the Hubble relationship. This is more than a mere similarity: the mathematical relationship is identical. The calculation can also be performed backwards. If spatial expansion reaches a certain rate at a distance of 46 billion light-years, we can determine the proportional expansion factor from it: H = (derivative of R) divided by R. With a distance increase of approximately 987,280 km/s, this yields: H approximately equal to 70 km/s/Mpc.

With this, we obtain the same value we previously used as a comparison value. This means: The Hubble relationship is mathematically compatible with our assumed radial expansion geometry. We have not yet proven that the assumed total energy of 140,000 observable universes physically causes this expansion rate. Nor have we proven that a central energy state produces the required internal spacetime evolution. Exceeding the speed of light during distance increase is not an immediate contradiction here. In an expanding spacetime, the distance between points far apart from each other can increase at a rate greater than the speed of light.

What We Still Want to Investigate

Our model has so far combined four central approaches:

- An extremely massive central antimatter core could explain part of the additional gravitational effect that we attribute to dark matter today.
- A spherically symmetric change in internal spacetime geometry could reproduce the observed Hubble relationship.
- The spatial separation of matter and antimatter could explain why both forms of matter did not completely annihilate each other after the initial formation phase.
- The central antimatter core could collapse under its own gravity into an extremely concentrated energy state that can potentially be interpreted within our particle hypothesis as a Universal Core Quark.

The next questions are:

- Can the assumed mass or energy distribution physically produce the required internal expansion?
- Can the gravitation of the core together with dynamic spacetime geometry explain the observed rotation curves?
- What influence does the proper rotation of galaxies have?
- What spatial distribution must outer matter possess?
- Can the assumed particle-antiparticle separation physically arise under extreme conditions?
- Can the antimatter core collapse into an extremely concentrated energy state?
- Can such a state cause a spherically symmetric internal spacetime evolution?
- Can the internal geometry expand while the outer Schwarzschild boundary remains unchanged?
- What energy-momentum distribution would be necessary for this?
- What effects would the existence of such a massive central core have on the large-scale distribution of matter and light?
- What observable differences arise compared to the established cosmological model?

In particular, we want to investigate whether the gravitation of the core and the dynamic evolution of internal spacetime influence each other and perhaps together explain the observed effects.

Summary of Previous Model Assumptions Preliminary parameters of the spherical model:

- Distance from core to Milky Way: 46 billion light-years
- Estimated core mass or core energy: 70,000 times the mass of the observable universe
- Assumed outer mass or outer energy: 70,000 times the mass of the observable universe
- Total mass or total energy equivalent: 140,000 times the mass of the observable universe
- Comparison value of Hubble constant: 70 km/s/Mpc

- Required instantaneous distance increase at 46 billion light-years: 3.55 billion km/h

The mass factors refer in each case to the estimated mass of the observable universe. They represent provisional model assumptions.

Composition of the system:

- Central antimatter core made of positrons, antiprotons, antineutrons, and possibly other antiparticles.
- Possibility of a further collapse into an extremely concentrated energy state.
- Hypothetical Universal Core Quark as a possible interpretation of this central state.
- Outer matter region made of ordinary matter.
- Spatial separation between both regions.
- An overall electrically neutral system is fundamentally possible with a corresponding charge balance.

Basic spacetime assumption:

- The entire universe is a black hole in this model.
- The outer Schwarzschild boundary initially remains unchanged.
- The internal spacetime can change dynamically.
- Expansion arises from an intrinsic change in spatial geometry.
- The increasing distance between core and outer matter is not necessarily attributable to ordinary movement of the core or matter through static space.

Basic particle assumption:

- All particles could be different states of a common photon-like energy foundation.
- Differences between particle types could arise from spin, polarization, state of motion, and other properties.
- Rest mass could emerge within this hypothesis from an altered motion structure.
- The Universal Core Quark could represent an extreme state of this energy foundation.

Our previous result:

The assumed spherically symmetric expansion geometry is mathematically compatible with the Hubble relationship. The first rough estimate also provides a preliminary magnitude for the mass of the central core. The extension of the core to an electrically neutralizable antimatter system eliminates a fundamental objection against the original idea of an exclusively positively charged positron core. With the introduction of the Universal Core Quark, another possible explanation arises: the original antimatter core could collapse under its own gravity into an extremely concentrated energy state whose energy continues to influence internal spacetime. The central new hypothesis states that the observed cosmic expansion does not arise primarily from the movement of outer matter away from the center, but rather from an intrinsic change in internal spacetime geometry. The outer size of the entire black hole could remain unchanged in the process, while the

spatial distances within its interior increase. All these approaches are currently independent of each other and not yet connected by a common mathematical solution. The spherical model is thus a speculative working hypothesis. It provides a coherent mechanical concept whose individual assumptions must now be tested mathematically and against astronomical observations.